Requirements

Objective

Implement a comprehensive function to analyze the data distribution in Data1.txt by calculating essential statistics and performing a sophisticated outlier analysis using the Interquartile Range (IQR). This analysis will include an "Outlier Scoring" system to classify different levels of outliers and explore how these affect various statistical measures. You will also use visualization to illustrate the data distribution and the effect of removing different levels of outliers.

Requirements

1. **Function Input**: A vector of numerical values loaded from Data1.txt.
2. **Function Output**:
    - **Basic Statistical Measures**:
        - Mean, Median, Standard Deviation, and Variance of the dataset before and after outlier removal.
    - **Quartile Values and IQR**:
        - Calculate the 1st quartile (Q1), 2nd quartile (Median), 3rd quartile (Q3), and Interquartile Range (IQR).
    - **Outlier Detection, Scoring, and Classification**:
        - Identify outliers using the IQR method:
            - *Outliers*: Values beyond Q1 - 1.5×IQR or Q3 + 1.5×IQR.
            - *Extreme Outliers*: Values beyond Q1 - 3×IQR or Q3 + 3×IQR.
        - Assign an **Outlier Score** for each identified outlier based on its distance from the IQR bounds:
            - For values above the upper bound: Outlier Score = (data point - Q3) / IQR
            - For values below the lower bound: Outlier Score = (Q1 - data point) / IQR
        - Classify outliers into the following levels based on scores:
            - **Moderate**: Outlier Score between 1.5 and 3
            - **High**: Outlier Score between 3 and 5
            - **Extreme**: Outlier Score greater than 5
    - **Impact Assessment of Outlier Levels**:
        - Calculate the mean, median, and standard deviation of the dataset after progressively removing each outlier level (first "Moderate," then "High," and finally "Extreme").

        - Summarize how these values change with the removal of each level, and provide a brief analysis of how different outlier types impact data distribution.

3. **Visualization**:
    - **Histogram and Box Plot**:
        - Plot the original data distribution as a histogram and box plot to visually show the range and distribution of data, including outliers.
    - **Outlier Removal Impact**:
        - After each level of outlier removal (Moderate, High, Extreme), create an updated histogram and box plot to visually show changes in the data's distribution.
    - **Line Plot for Statistics**:
        - Use a line plot to show the change in mean, median, and standard deviation as each outlier level is removed, enabling visual analysis of how each step impacts overall statistics.
4. **Deliverables**:
    - **Python Code**: A script implementing the function as specified above, complete with comments and organized code structure.
    - **Analysis Report**:
        - A summary of the calculated statistics (mean, median, standard deviation, variance, quartiles) before and after each level of outlier removal.
        - An itemized list of identified outliers with their scores and classifications.
        - A table or visualization showing changes in mean, median, and standard deviation as each outlier level is removed.
        - A brief discussion on the impact of different levels of outliers on the dataset and possible real-world reasons for the presence of outliers (e.g., data entry errors, unique events, or sensor faults).
        - Visualizations (histogram, box plot, and line plot) with brief captions explaining each.
